
DATA ANNOTATION - FINANCIAL DOCUMENT CLASSIFICATION & ENTITY EXTRACTION ANNOTATION FRAMEWORK

Category: Data Annotation | Natural Language Processing & AI Training Data

Timeline: 15 Days | AI-Accelerated Innovation & Research Project

Target: Undergraduate / MBA / Engineering Students

TASK:

Design and deliver a complete financial document classification and entity extraction annotation framework for a fictional document intelligence startup - FinSight AI - whose OCR and extraction model sits at 72% accuracy and must reach 95%+ to close its Series A funding round. As Data Annotation Lead, you must build the entire annotation infrastructure from scratch across a 200-document corpus spanning eight categories - bank statements, commercial invoices, income tax returns, loan agreements, insurance policies, regulatory filings, payment receipts, and financial statements. You must design a labelling schema covering a minimum of 22 entity types including monetary amounts, regulatory IDs, clause references, computation fields, and tabular data structures; author a comprehensive annotation guideline with edge case decision rules for each document category; produce a gold-standard annotated dataset; and demonstrate through rigorous inter-annotator agreement metrics that your framework generates consistent, reproducible labels - all to a standard where a downstream Model Training Team can directly use your output to train a production-grade financial document AI without schema redesign.

PART A: TRAINING MATERIAL & LEARNING RESOURCES

PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

PART C: CASE STUDIES & REAL-WORLD APPLICATIONS

PART D: 15-DAY PROJECT METHODOLOGY & DELIVERABLES

PART E: NO-CODE TOOLS & PLATFORMS GUIDE

PART F: SUBMISSION PROTOCOL

Important Note:

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EXECUTIVE SUMMARY

This project positions students as Data Annotation Lead at a fictional document intelligence startup - FinSight AI - requiring them to master the convergence of financial document processing expertise, entity taxonomy design, annotation quality governance, and NLP data engineering. Unlike general annotation projects that supply a pre-built schema and ask students to apply labels, this project operates at the architectural layer - demanding fluency in labelling schema design, inter-annotator agreement measurement, and edge case resolution methodology on one side, and the specific structural complexity of eight distinct financial document types on the other - because a misclassified entity in a financial document is not merely a data quality issue; it is a compliance violation, a failed audit, or a material financial loss waiting to propagate through the downstream model's predictions at scale.

The core challenge students must solve: How can a single annotation schema impose consistent, reproducible labels across 200 financial documents that span the full complexity spectrum - from relatively structured payment receipts to the very high complexity of loan agreements with nested clause hierarchies and variable rate schedules, regulatory filings with XBRL tags and multi-entity disclosures, and financial statements with IFRS versus GAAP variations and auditor notes - across 22 entity types whose boundaries are frequently ambiguous, whose formats vary by jurisdiction and institution, and whose correct interpretation requires genuine financial domain knowledge to navigate? The answer lies in a four-layer annotation quality architecture built across taxonomy design, guideline authorship, calibration, and measurement - where every entity type definition is stress-tested against the hardest edge cases before a single document is labelled, validation rules are specified per entity type to catch format violations before they enter the dataset, and inter-annotator agreement is measured separately by document category to identify where schema revision is needed rather than where annotators need retraining.

Students will design and validate a complete annotation pipeline across five progressive workstreams: a 22+ entity type taxonomy with definitions, data types, validation rules, and explicit decision rules for boundary cases including multi-currency documents, ambiguous date formats, PII masking requirements for account numbers and contact information, and the distinction between interest rate and percentage entity types; a comprehensive annotation guideline document with worked examples for each of the eight document categories and a minimum of one calibration example per edge case identified during taxonomy design; a gold-standard annotated dataset covering the full 200-document corpus with all entity spans, types, and structured attribute values populated in a consistent output format; an inter-annotator agreement analysis computing Cohen's Kappa and Krippendorff's Alpha separately at the entity type and span boundary levels, with disagreement analysis broken down by document category and entity type; and a quality assurance report documenting schema revision history, adjudication decisions, systematic annotator error patterns, and a before-and-after accuracy projection linking annotation quality improvements to the target 95%+ model accuracy threshold. Every entity annotation must include a span boundary, a type from the defined taxonomy, and any required structured attributes, and every guideline decision must be traceable to a specific validation rule or edge case resolution.

Financial document AI systems generate over \$2.7 billion in annual revenue globally, with accuracy requirements so stringent that a single entity misclassification can trigger regulatory violations or material financial reporting errors - yet the discipline of designing annotation frameworks that produce the label quality these systems require, rather than simply applying labels to existing schemas, remains one of the most overlooked skill areas in AI and data science education. This project closes it directly, equipping students with the taxonomy design rigour, financial document domain expertise, and annotation quality governance capability that data annotation specialist roles at AI companies, document intelligence platforms, and financial technology firms demand.

PART A: TRAINING MATERIAL & LEARNING RESOURCES

SECTION A1 - EXECUTIVE OVERVIEW & PURPOSE

This document constitutes the definitive, binding assessment framework for Project 1A within the Data Annotator career track at Zetheta Algorithms Private Limited. It serves as the single authoritative reference for all interns undertaking the Financial Document Classification & Entity Extraction Annotation Framework project. Every deliverable, evaluation criterion, timeline milestone, and quality threshold specified herein is non-negotiable and contractually binding upon acceptance of the internship.

The document has been architected to operate at the highest levels of professional rigour, mirroring the expectations that would be placed upon a junior data annotation specialist at a Tier-1 AI company (Google DeepMind, OpenAI, Anthropic, Scale AI, Snorkel AI, or equivalent). Interns are expected to read this document in its entirety before commencing work and to reference it continuously throughout the seven-day assessment period. Failure to read and understand any section does not constitute grounds for leniency in assessment.

SECTION A2 - THE STRATEGIC IMPERATIVE OF DATA ANNOTATION IN FINANCIAL AI

In the current AI landscape, data annotation quality is the single most significant determinant of downstream model performance. Research consistently demonstrates that a 10% improvement in training data quality yields a 15–25% improvement in model accuracy, whereas equivalent investments in model architecture changes yield only 3–8% gains. This asymmetry makes data annotation a strategic discipline of paramount importance, not a mechanical task.

Financial document processing represents one of the highest-value annotation domains, with commercially deployed document AI systems generating USD 2.7 billion in annual revenue globally as of 2025. The accuracy requirements in financial document processing are extraordinarily stringent: a single misclassified entity (e.g., confusing a payment date with an invoice date) can trigger compliance violations, incorrect financial reporting, failed regulatory audits, or material financial losses running into millions.

This project has been engineered to compress what would normally be a 3–6 month onboarding programme into an intensive 7-day sprint, forcing interns to make rapid, consequential decisions while maintaining quality standards that meet or exceed industry benchmarks. The compression ratio is deliberate: it separates candidates who can perform under pressure from those who cannot.

SECTION A3 - GAMIFIED SIMULATION CONTEXT

YOUR MISSION BRIEFING

You have been appointed as the Data Annotation Lead at FinSight AI, a document intelligence startup whose OCR + extraction model currently achieves only 72% accuracy on financial document processing tasks. The company needs to reach 95%+ accuracy to achieve commercial viability and close its Series A funding round.

You have received 200 sample documents spanning 8 categories. Your mandate is to build the complete annotation infrastructure from scratch: design the labelling schema, author a comprehensive annotation guideline, produce a gold-standard dataset, and demonstrate through rigorous inter-annotator agreement metrics that your framework produces consistent, high-quality annotations.

A Model Training Team will evaluate whether data annotated using your framework actually improves model performance on a held-out test set. Your work directly determines whether FinSight AI survives or fails.



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SECTION A4 - DOCUMENT CORPUS SPECIFICATION

The project requires interns to work with a corpus of 200 financial documents spanning eight distinct categories. Each category presents unique annotation challenges, and interns must demonstrate mastery of all categories. The following table specifies the exact corpus composition and complexity ratings:

#	Document Category	Count	Complexity	Key Challenges
1	Bank Statements	30	Medium	Multi-account layouts, running balances, variable transaction descriptions, multi-currency entries
2	Commercial Invoices	30	Medium	Tax calculations (GST/VAT/Sales Tax), discount structures, line-item hierarchies, partial payments
3	Income Tax Returns	25	High	Multi-schedule forms, cross-referenced sections, computed vs. reported values, jurisdiction variations
4	Loan Agreements	25	Very High	Legal terminology, nested clause structures, variable rate schedules, guarantor information, collateral descriptions
5	Insurance Policies	20	High	Policy-specific terminology, exclusion clauses, coverage tables, endorsement riders, premium schedules
6	Regulatory Filings	20	Very High	Jurisdiction-specific formats, XBRL tags, multi-entity disclosures, footnote references, restated figures
7	Payment Receipts	25	Low-Medium	Abbreviated descriptions, partial information, multiple payment methods, tip/gratuuity handling
8	Financial Statements	25	Very High	Consolidated vs. standalone figures, prior period comparisons, IFRS vs. GAAP variations, auditor notes

SECTION A5 - ENTITY TAXONOMY - COMPLETE SPECIFICATION

Interns must design an annotation schema that captures a minimum of 22 entity types across the document corpus. The following master taxonomy provides the required entity types, along with their definitions, data types, and validation rules. Interns may extend this taxonomy (and are encouraged to do so for bonus points) but may not reduce it under any circumstances.

#	Entity Type	Definition	Data Type	Validation / Edge Case Notes
1	MONETARY_AMOUNT	Any numerical value representing money, including currency symbol/code	Numeric + Currency	Must capture currency; handle multi-currency in single doc
2	DATE	Calendar dates in any format (DD/MM/YYYY, Month DD YYYY, fiscal dates)	Date	Normalize to ISO 8601; flag ambiguous formats (01/02/2024)
3	ACCOUNT_NUMBER	Bank accounts, loan accounts, policy numbers, folio numbers	Alphanumeric	Mask PII in output; validate format per institution
4	PARTY_NAME	Names of individuals, companies, trusts, or government entities	String	Distinguish payer vs. payee vs. guarantor vs. beneficiary
5	REGULATOR_Y_ID	PAN, GSTIN, TIN, EIN, LEI, ISIN, or equivalent identifiers	Alphanumeric	Validate checksum where applicable (PAN, GSTIN)
6	ADDRESS	Physical addresses including street, city, state, postal code, country	Structured	Parse into sub-components; handle incomplete addresses
7	INTEREST_RATE	Annual, monthly, or periodic interest rates	Numeric + Period	Distinguish APR vs. flat rate vs. compounding rate

#	Entity Type	Definition	Data Type	Validation / Edge Case Notes
8	DURATION	Time periods (loan tenure, policy term, notice period)	Numeric + Unit	Normalize to standard units (months/days)
9	PERCENTAGE	Non-interest percentages (tax rates, discount rates, ownership %)	Numeric	Distinguish from interest rates via context
10	DOCUMENT_ID	Invoice numbers, filing reference, case numbers, policy IDs	Alphanumeric	Cross-reference within and across documents
11	CLAUSE_REFERENCE	References to legal sections, schedules, annexures	String	Maintain hierarchical structure (Section 3.2(a)(ii))
12	TRANSACTION_TYPE	Debit, credit, transfer, reversal, adjustment, refund	Enum	Standardize variations (DR/Debit/D all map to DEBIT)
13	INSTITUTION_NAME	Banks, insurance companies, regulatory bodies, exchanges	String	Distinguish from general PARTY_NAME; use canonical names
14	PRODUCT_NAME	Financial products (Fixed Deposit, Term Insurance, Mutual Fund)	String	Map to standardized product categories
15	STATUS	Document/transaction status (Active, Lapsed, Pending, Approved)	Enum	Capture temporal dimension (status as of date X)
16	CONTACT_INFO	Phone numbers, email addresses, fax numbers, URLs	Structured	PII handling required; validate format

#	Entity Type	Definition	Data Type	Validation / Edge Case Notes
17	SIGNATURE_BLOCK	Signatory names, designations, dates of signing	Structured	Link to PARTY_NAME entity; capture authority level
18	TABLE_DATA	Structured tabular information (amortization schedules, coverage tables)	Table	Preserve row/column relationships; handle merged cells
19	COMPUTATION	Calculated fields (total = subtotal + tax, balance = opening + credits - debits)	Formula	Capture the computation logic, not just the result
20	FOOTNOTE_REF	References to footnotes, endnotes, or explanatory notes	Reference	Link footnote content to referencing text
21	CURRENCY_CODE	ISO 4217 currency codes or symbols	Enum	Normalize symbols to ISO codes; handle ambiguity (\$ = USD/SGD/AUD)
22	FISCAL_PERIOD	Financial year, quarter, assessment year references	Period	Distinguish calendar year from fiscal year

SECTION A6 - DOCUMENT CLASSIFICATION SCHEMA - HIERARCHICAL SPECIFICATION

Beyond the primary 8-category classification, interns must implement a hierarchical classification scheme with secondary and tertiary labels. The following multi-level schema is mandatory and must be fully implemented in the annotation schema.

A6.1 Level 1: Primary Document Type

The eight categories specified in Section A.5 form the primary classification level. Each document must receive exactly one primary label. Misclassification at this level is considered a critical error.

A6.2 Level 2: Document Subtype

Each primary category must be subdivided into at least three subtypes. The following table specifies the minimum required subtypes:

Primary Category	Required Subtypes (Minimum 3)
Bank Statements	Savings Account Statement, Current Account Statement, Credit Card Statement, Fixed Deposit Statement, Demat Account Statement
Commercial Invoices	Tax Invoice, Proforma Invoice, Debit Note, Credit Note, Self-Invoice (Reverse Charge)
Income Tax Returns	ITR-1 (Sahaj), ITR-2, ITR-3, ITR-4 (Sugam), ITR-5 (Firms), ITR-6 (Companies), ITR-7 (Trusts)
Loan Agreements	Personal Loan, Home Loan / Mortgage, Vehicle Loan, Business Loan, Overdraft Facility, Line of Credit
Insurance Policies	Term Life, Health / Mediclaim, Motor (Comprehensive/TP), Property / Fire, Professional Indemnity, Marine Cargo
Regulatory Filings	Annual Return (MCA), GST Return (GSTR-1/3B/9), TDS Return (26Q/27Q), SEBI Filing, RBI Reporting
Payment Receipts	Cash Receipt, Online Payment Confirmation, Cheque Receipt, Demand Draft Receipt, UPI Payment Confirmation
Financial Statements	Balance Sheet, Profit & Loss Statement, Cash Flow Statement, Notes to Accounts, Director Report Extract

A6.3 Level 3: Document Quality & Condition Tags

Every document must also receive metadata tags describing its quality and condition. These tags affect annotation difficulty and must be tracked for efficiency analysis. Multiple tags may apply to a single document.

Quality Tag	Definition
CLEAN_DIGITAL	Natively digital document with clear text and formatting
OCR_HIGH_QUALITY	Scanned document with OCR confidence > 95%
OCR_DEGRADED	Scanned document with OCR confidence 80–95%, some character errors
OCR_NOISY	Scanned document with OCR confidence < 80%, significant errors
MULTI_LANGUAGE	Document contains text in two or more languages
MULTI_CURRENCY	Document contains amounts in two or more currencies
HANDWRITTEN_ELEMENTS	Contains handwritten notes, signatures, or annotations
REDACTED	Contains intentionally obscured or redacted information
TEMPLATE_MISMATCH	Document format does not match standard templates
INCOMPLETE	Document is truncated, missing pages, or partially illegible

SECTION A7 - ENTITY RELATIONSHIP MODEL - REQUIRED CONNECTIONS

The annotation schema must capture not only individual entities but the semantic relationships between them. The following relationship types are mandatory and must be implemented in the schema's relationships section. Failure to model these relationships will result in significant point deductions under the Entity Relationships criterion.

Relationship Type	Source Entity	Target Entity	Description	Example
HOLDER_OF_ACCOUNT	PARTY_NAME	ACCOUNT_NUMBER	Person or entity that owns or operates the account	Rajesh Kumar > Account 1234567890
PAYER_OF	PARTY_NAME	MONETARY_AMOUNT	Entity making a payment	ABC Corp pays Rs. 50,000
PAYEE_OF	PARTY_NAME	MONETARY_AMOUNT	Entity receiving a payment	XYZ Ltd receives Rs. 50,000
GUARANTOR_FOR	PARTY_NAME	PARTY_NAME	Entity guaranteeing another entity's obligation	Mr. Singh guarantees Ms. Patel's loan
ISSUED_BY	DOCUMENT_ID	INSTITUTION_NAME	Document issued by an institution	Invoice INV-2024-001 issued by HDFC Bank
COVERS_PERIOD	DOCUMENT_ID	FISCAL_PERIOD	Document covers a specific fiscal period	Balance Sheet covers FY 2023-24
COMPUTED_FROM	MONETARY_AMOUNT	COMPUTATION	Amount derived from a calculation	Total Rs. 11,800 = Subtotal + GST
REFERENCES_CLAUSE	MONETARY_AMOUNT	CLAUSE_REFERENCE	Amount defined in a specific clause	Penalty per Section 12.3(a)

Relationship Type	Source Entity	Target Entity	Description	Example
DENOMINATED_IN	MONETARY_AMOUNT	CURRENCY_CODE	Currency of a monetary amount	Rs. 45,230 denominated in INR
SIGNED_BY	SIGNATURE_BLOCK	PARTY_NAME	Document signed by a specific party	Signed by Director Amit Sharma
EFFECTIVE_ON	STATUS	DATE	Status effective as of a specific date	Policy Active as of 15-Mar-2024
CONTAINS_TABLE	DOCUMENT_ID	TABLE_DATA	Document contains structured tabular data	Loan schedule in Table 4
FOOTNOTE_FOR	FOOTNOTE_REF	MONETARY_AMOUNT	Footnote explaining a financial figure	Note 7 explains Revenue figure
SUBSIDIARY_OF	PARTY_NAME	PARTY_NAME	Entity is a subsidiary of another	ABC India is subsidiary of ABC Global

SECTION A8 - DOCUMENT CORPUS SOURCING GUIDELINES

Interns are responsible for sourcing or generating the 200-document corpus. This is itself an assessed skill - the quality and realism of the corpus directly affects the quality of all downstream deliverables. The following guidelines must be followed:

A8.1 Acceptable Sources

- **Synthetic Document Generation:** Create realistic financial documents using templates in Python (reportlab, fpdf2), LaTeX, or document generation APIs. This is the preferred method as it guarantees no PII issues.
- **Public Datasets:** Use publicly available financial document datasets from Kaggle, HuggingFace Datasets, or academic repositories. Cite all sources in the README.
- **Government/Regulatory Sources:** Use publicly available regulatory filings from MCA, SEBI EDGAR, or equivalent public databases (these are already public records).
- **Template Modification:** Start with publicly available document templates and populate them with synthetic data to create realistic but fictional documents.

A8.2 Corpus Quality Requirements

Requirement	Specification	Rationale
Category Distribution	Exact counts as specified in Section A.5 (30/30/25/25/20/20/25/25)	Ensures balanced exposure to all document types
Complexity Distribution	At least 20% of documents must be rated difficulty 4 or 5	Tests ability to handle challenging edge cases
Quality Tag Coverage	Each quality tag (Section A.7.3) must appear in at least 5 documents	Ensures experience with all quality conditions
Multi-Language Inclusion	At least 10 documents must include multi-language content	Tests multi-language annotation capability
Multi-Currency Inclusion	At least 15 documents must include multi-currency amounts	Tests currency disambiguation skills
OCR Simulation	At least 25 documents must simulate OCR artifacts (degraded/noisy)	Tests OCR error handling and correction

Requirement	Specification	Rationale
Template Variety	No more than 3 documents may use the same visual template per category	Prevents overfitting to specific layouts
Realistic Content	All financial figures must be internally consistent (or deliberately inconsistent for edge cases)	Tests attention to detail and computation verification



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PART B: GAMIFIED SIMULATION BUSINESS CONCEPT

Note on the Simulation Concept: The gamified simulation concept presented in this section is provided as a reference framework. Students may choose to follow this concept or design their own original gamified simulation - with a different name, structure, narrative, or game mechanics - as long as it meets the assessment criteria outlined in Part F.

The following matrix provides a high-level overview of all seven deliverables, their relative weights, key quality indicators, and interdependencies. Refer to subsequent sub-sections for exhaustive specifications.

Deliverable Name	Format	Key Quality Indicator
Annotation Schema	JSON Schema (annotation_schema.json)	22+ entity types with validation rules and relationships
Annotation Guideline	PDF + Source (30+ pages)	New annotator achieves Kappa ≥ 0.80 within 20 docs
Gold-Standard Dataset	JSONL (200 documents)	98%+ entity extraction accuracy; 100% schema conformance
IAA Report	PDF report	Cohen's Kappa ≥ 0.85 with confidence intervals
Edge Case Taxonomy	Markdown (30+ cases)	All 9 categories covered with resolution reasoning
QA Workflow Spec	Markdown	Five-stage pipeline with quality gates and escalation
Efficiency Metrics	Markdown	Learning curve showing improvement over 7 days

B.1 Deliverable D1: Annotation Schema

B.1.1 Requirements

- Format: JSON Schema file (annotation_schema.json) conforming to JSON Schema Draft 2020-12
- Minimum Entity Types: 22 entity types as specified in Section A.6, plus any extensions
- Document Types: 8 primary categories, 3+ subtypes per category, 10 quality tags
- Relationships: Schema must define relationships between entities (e.g., MONETARY_AMOUNT linked to CURRENCY_CODE)
- Validation Rules: Each entity type must include regex patterns, allowed value ranges, or enum constraints
- Annotation Attributes: Each entity annotation must support confidence score (0.0–1.0), source location (page/bounding box), OCR quality flag, and annotator notes

B.1.2 Schema Structure Requirements

The annotation schema must contain the following top-level sections:

Section	Required Fields
document_metadata	Document ID, source file path, creation date, last modified, assigned annotator, review status
classification	Primary type, subtype, quality tags (multi-label), confidence scores per level
entities	Array of entity objects with: entity_id, entity_type, value, normalized_value, source_text, page_number, bounding_box, confidence, linked_entities, annotator_notes
relationships	Entity-to-entity relationships (e.g., payer_of, guarantor_for, references_clause)
quality_metadata	OCR confidence per page, annotation difficulty rating, time_to_annotate, review_comments
version_control	Schema version, annotation guideline version, annotator tool version

B.2 Deliverable D2: Annotation Guideline Document

B.2.1 Minimum Content Requirements

The annotation guideline must be a minimum of 30 pages (A4, 11pt font, single-spaced equivalent) and must contain all of the following sections:

#	Section	Min Pages	Required Content
1	Introduction & Purpose	2–3	Scope of the guideline, target audience, how to use the document, version history
2	Document Classification Guide	3–4	Decision tree for primary classification, subtype identification criteria, quality tag assignment rules with examples
3	Entity Type Definitions	8–10	Comprehensive definition of each entity type with 2–3 positive examples and 1–2 negative examples per type (minimum 50 labelled examples total)
4	Annotation Conventions	2–3	Span selection rules (minimal span vs. maximal span), nested entity handling, discontinuous entity handling, overlapping entities
5	Edge Case Compendium	5–6	Minimum 30 documented edge cases with resolution guidelines, categorized by difficulty level and entity type
6	Document-Specific Instructions	4–5	Specific annotation instructions for each of the 8 document categories, highlighting category-specific challenges
7	Quality Tag Assignment Guide	2–3	How to assess and assign each quality tag, with example images/descriptions of each quality level
8	Common Errors & Anti-Patterns	2–3	Catalogue of frequently observed annotation errors, why they occur, and how to avoid them
9	Appendices	2–3	Abbreviation glossary, regex reference for entity validation, decision flowcharts

B.2.2 Labelled Example Requirements

The guideline must contain a minimum of 50 labelled examples distributed across entity types and document categories:

- At least 2 positive examples per entity type (22 types $\times$ 2 = 44 minimum)
- At least 1 negative example (what NOT to annotate) per entity type (22 additional)
- At least 3 complex multi-entity examples showing entity relationships
- At least 5 edge case examples with detailed resolution explanations
- Each example must include: source text snippet, entity span highlighted, entity type label, normalized value, annotator reasoning



B.3 Deliverable D3: Gold-Standard Annotated Dataset

B.3.1 Requirements

- **Corpus Size:** 200 documents across all 8 categories as specified in Section A.5
- **Annotation Format:** JSON Lines (.jsonl) format, one document per line, conforming to the annotation schema (D1)
- **Classification:** Every document must have primary type, subtype, and quality tags assigned
- **Entity Extraction:** Every identifiable entity in every document must be annotated (no partial annotation)
- **Normalized Values:** All entity values must include both raw extracted text and normalized form
- **Quality Threshold:** Gold-standard annotations must be reviewed at least once, with all disagreements resolved

B.3.2 Annotation Completeness Matrix

Category	Check	Mandatory	Notes
Document Classification	Primary type assigned	YES	Auto-reject if missing
Document Classification	Subtype assigned	YES	Auto-reject if missing
Document Classification	Quality tags assigned (at least 1)	YES	Default to CLEAN_DIGITAL if truly pristine
Entity Extraction	All monetary amounts annotated	YES	Verify against document totals where possible
Entity Extraction	All dates annotated and normalized	YES	ISO 8601 format mandatory
Entity Extraction	All party names identified	YES	Includes abbreviated/partial names
Entity Extraction	All regulatory IDs captured	YES	Validate format (PAN: XXXXX0000X)

Category	Check	Mandatory	Notes
Entity Extraction	All account numbers captured	YES	PII masking applied
Entity Extraction	All table data preserved	YES	Row/column structure maintained
Quality Metadata	OCR confidence recorded per page	YES	May be 1.0 for digital-native docs
Quality Metadata	Annotation difficulty rating assigned	YES	Scale 1–5
Quality Metadata	Time-to-annotate recorded	YES	In seconds

B.4 Deliverable D4: Inter-Annotator Agreement Report

B.4.1 Study Design Requirements

The inter-annotator agreement (IAA) study must demonstrate that the annotation guideline (D2) produces consistent annotations across different annotators. Even though the intern may be working alone, they must simulate a multi-annotator environment using one of the following methods:

1. Method A (Preferred): Recruit 2–3 peers to annotate a subset of 30 documents independently, then compute agreement metrics.
2. Method B: Re-annotate 30 documents yourself after a 48-hour gap, then compute self-agreement metrics (test-retest reliability).
3. Method C: Use an LLM-based annotator (GPT-4, Claude) to annotate 30 documents using your guideline, then compare against your gold-standard annotations.

B.4.2 Required Metrics

Metric	Formula/Method	Min Threshold	Reporting Requirement
Cohen's Kappa (Classification)	$Kappa = (Po - Pe) / (1 - Pe)$	≥ 0.85	Per-category and overall; confusion matrix required
Fleiss' Kappa (if 3+ annotators)	Multi-rater extension of Cohen's Kappa	≥ 0.80	Overall and per-entity-type
Entity-Level F1 Score	Precision/Recall/F1 per entity type	≥ 0.85 macro-F1	Breakdown by entity type; highlight lowest-performing types
Exact Match Ratio	Proportion of entities with identical spans	≥ 0.75	Compare strict vs. relaxed span matching
Span Overlap (Token-Level)	Jaccard similarity of annotated spans	≥ 0.80	Report mean and distribution
Boundary Agreement	% of entities where start and end positions agree	≥ 0.70	Separate analysis for each entity type

B.4.3 Report Structure

The IAA report must include the following sections:

- Executive summary with headline Kappa scores and pass/fail determination
- Methodology description (which method was used, how annotators were briefed, subset selection criteria)
- Per-category classification agreement with confusion matrices
- Per-entity-type extraction agreement with precision/recall/F1 tables
- Error analysis identifying the top 10 sources of disagreement
- Guideline revision recommendations based on disagreement patterns
- Statistical significance testing (p-values for Kappa scores; bootstrap confidence intervals)
- Raw agreement data as appendix (to verify calculations)



B.5 Deliverable D5: Edge Case Taxonomy

B.5.1 Requirements

The edge case taxonomy must document a minimum of 30 edge cases encountered during annotation. Each edge case must be a genuine challenge that required human judgement to resolve, not a trivial variation.

B.5.2 Edge Case Documentation Format

Field	Description
Edge Case ID	Unique identifier (EC-001, EC-002, ...)
Category	Classification, Entity Extraction, Normalization, Relationship, or Quality Assessment
Difficulty Level	Level 1 (Common), Level 2 (Uncommon), Level 3 (Rare/Ambiguous)
Document Category	Which document type(s) this edge case applies to
Description	Clear description of the ambiguity or challenge
Example	Actual text snippet demonstrating the edge case
Resolution	How the edge case should be annotated and why
Alternative Resolutions Considered	What other approaches were considered and why they were rejected
Frequency	Estimated frequency in the corpus (X out of 200 documents)
Impact on Model	How incorrect handling would affect downstream model performance

B.5.3 Required Edge Case Categories

Category	Min Count	Examples of Edge Cases
Multi-Currency Handling	5+	Documents containing amounts in multiple currencies; conversion rate ambiguity; currency symbol disambiguation (\$ = USD/SGD/AUD)
OCR Noise & Errors	4+	Character confusion (0/O, 1/I), truncated text, misaligned columns, garbled special characters
Multi-Language Content	3+	Code-switching within documents, non-Latin script handling, transliterated names
Ambiguous Dates	3+	DD/MM vs. MM/DD ambiguity, fiscal year vs. calendar year, relative dates
Nested / Overlapping Entities	3+	Address containing institution name, amount containing date reference, clause with multiple entity types
Computed vs. Stated Values	3+	Discrepancies between stated totals and computed sums, rounding differences
Template Variations	3+	Same document type from different issuers with incompatible layouts
Legal / Regulatory Ambiguity	3+	Clause references spanning multiple sections, conditional terms, undefined acronyms
PII and Redaction	3+	Partially masked account numbers, redacted names requiring inference, synthetic data artifacts

B.6 Deliverable D6: Quality Assurance Workflow Specification

The QA workflow specification must describe the end-to-end process for ensuring annotation quality across the entire 200-document corpus. It must be detailed enough that a project manager could implement it for a team of 10 annotators. The specification must include:

1. Annotation pipeline flowchart (pre-annotation, annotation, review, adjudication, final QA)
2. Role definitions (Annotator, Reviewer, Adjudicator, QA Lead) with responsibilities
3. Quality gates at each stage with pass/fail criteria
4. Sampling strategy for review (what percentage of documents undergo full review)
5. Escalation procedures for disagreements that cannot be resolved at the annotator level
6. Feedback loop mechanism for updating the annotation guideline based on QA findings
7. Tooling requirements (Label Studio configuration, review queue management, metrics dashboard)



B.7 Deliverable D7: Annotation Efficiency Metrics

Interns must track and report their own annotation efficiency throughout the project:

Metric	Definition	Granularity	Expected Outcome
Time per Document	Total annotation time including entity extraction and review	Per document	Report mean, median, P90, P95 by document category
Entities per Hour	Number of entities annotated per hour of work	Per session	Should show learning curve (improvement over time)
Error Rate Over Time	Proportion of annotations corrected during review	Per day	Should decrease from Day 1 to Day 7
Annotation Throughput	Documents fully annotated per hour	Per day	Report separate throughput by complexity tier
Review Time Ratio	Time spent on review vs. initial annotation	Per day	Typically 0.3–0.5x of annotation time
Edge Case Encounter Rate	Number of new edge cases discovered per 10 documents	Cumulative	Should plateau as taxonomy matures

PART C: CASE STUDIES AND REAL-WORLD APPLICATIONS

CASE STUDY 1: HOW SCALE AI BUILT THE ANNOTATION INFRASTRUCTURE BEHIND FINANCIAL DOCUMENT AI

Background

Scale AI, one of the world's largest data annotation platforms, was contracted by a major US financial institution to build training data for an automated document processing system handling over 40 million financial documents annually - spanning loan agreements, regulatory filings, tax forms, and bank statements. The bank's existing manual review process cost \$180 million per year in labour and introduced a 3-5 day processing lag that was creating customer attrition. The target was a model that could process each document in under 4 seconds with 96%+ field-level extraction accuracy.

The Annotation Challenge

Scale AI's team discovered within the first two weeks that the fundamental bottleneck was not model architecture but annotation consistency. Their initial taxonomy had 18 entity types with loosely defined boundaries. When the same loan agreement was annotated by three different annotators, entity span agreement was only 61% - well below the 80%+ threshold required for reliable model training. The root causes were immediately identifiable: the MONETARY_AMOUNT entity type had no validation rule distinguishing principal amounts from penalty amounts from escrow balances; the DATE entity type had no normalisation standard, so annotators were inconsistently capturing fiscal dates versus calendar dates; and the CLAUSE_REFERENCE entity had no hierarchical structure specification, producing flat references where nested clause structures existed.

The Remediation

Scale AI paused annotation entirely and invested two weeks in schema redesign. They introduced sub-type attributes for every entity type that had been producing disagreements, wrote 47 edge case decision rules derived directly from the disagreement logs, and created a 120-item calibration test that every annotator had to pass at 90%+ before returning to live annotation. They also implemented a stratified sampling approach - pulling 5% of every annotator's daily output for cross-annotator verification, with automatic flag and adjudication for any disagreement.

Results & Simulation Connection

After schema redesign, inter-annotator agreement rose from 61% to 88% Cohen's Kappa on entity spans and 84% on entity types. Model accuracy on the held-out test set improved from 71% to 94.3% - achieving the commercial viability threshold. The bank's processing lag dropped from 3-5 days to 6 hours. Students undertaking this project face an identical structural challenge: the 22-entity taxonomy, the eight document categories with varying complexity ratings, and the inter-annotator agreement targets are all calibrated to simulate the exact conditions Scale AI encountered. The lesson is direct - annotation schema quality is not a prerequisite to the real work; it is the real work.

CASE STUDY 2: THE GSTIN MISCLASSIFICATION THAT COST AN INDIAN FINTECH ₹4.2 CRORE

Background

In 2022, a Bengaluru-based lending fintech deployed an ML-powered document verification system trained on an internally annotated dataset of 15,000 commercial invoices and GST filings. The system was designed to automatically extract GSTIN numbers, verify them against the GSTN portal, and flag discrepancies for manual review as part of the KYC process for SME loan applicants. The model went live without a formal annotation quality audit.

The Failure

Eight months after deployment, an internal audit revealed that the model had been systematically misclassifying PAN numbers as GSTINs in documents where both appeared in close proximity - a layout pattern common in GST invoices issued by sole proprietors, where the PAN-based GSTIN and the underlying PAN number appear within two lines of each other. The annotation team had not defined a validation rule distinguishing the two entity types, and individual annotators had resolved the ambiguity inconsistently across the training corpus. The model learned the majority pattern, which happened to be wrong in roughly 23% of sole proprietor documents.

The consequence: 1,847 loan applications were incorrectly verified over eight months. Of these, 312 involved fraudulent applicants whose mismatched PAN-GSTIN combinations should have triggered rejection but were approved. Total non-performing loan exposure attributable to the annotation failure: ₹4.2 crore. The company also faced a show-cause notice from the RBI for KYC process failures.

The Root Cause & Lesson

A post-mortem identified three annotation failures that compounded each other: no checksum validation rule for GSTIN (a valid GSTIN has a deterministic relationship to the underlying PAN that any validation rule would have encoded), no edge case decision rule for the sole proprietor document layout, and no stratified sampling of sole proprietor invoices during quality review - meaning the systematic error pattern was invisible in aggregate agreement statistics. The inter-annotator agreement score for the REGULATORY_ID entity type was a respectable 0.79 Kappa overall, but when disaggregated by document sub-type, sole proprietor invoices showed only 0.51 - a signal that was never examined.

Students in this project must navigate the same risk: the REGULATORY_ID entity type in the taxonomy carries an explicit note to validate checksum where applicable for PAN and GSTIN, and the document corpus includes regulatory filings and commercial invoices where these entities co-occur. The case study's lesson is that aggregate inter-annotator agreement scores can mask catastrophic sub-category failures - which is precisely why this project requires agreement analysis broken down by both document category and entity type, not reported only at the corpus level.

PART D: PROJECT METHODOLOGY AND DELIVERABLES

Day	Focus Area	Key Activities	Milestone Checkpoints
Day 1-3	Foundation & Schema Design	Read this document in full. Set up development environment (GitHub repository, Label Studio instance, documentation tools). Design and implement the annotation schema (D1). Begin drafting entity type definitions.	GitHub repo created with proper structure; Draft annotation_schema.json committed; Label Studio project initialized
Day 4-5	Annotation Guideline Development	Complete the annotation guideline core sections: document classification guide, entity type definitions with examples, annotation conventions. Begin annotating the first 30 documents to test the schema.	Guideline Sections 1–4 drafted (15+ pages); 30 documents annotated as pilot batch; Schema validated against pilot
Day 6-8	Edge Cases & Guideline Refinement	Continue annotation (target: 80 cumulative). Document edge cases as encountered. Refine the guideline based on edge cases. Complete document-specific instructions.	80 documents annotated; 15+ edge cases documented; Guideline Sections 5–6 completed; Schema revisions committed
Day 9	Annotation Sprint & Quality Gates	Annotation sprint (target: 140 cumulative). Implement the QA workflow. Begin self-review of completed annotations. Continue building edge case taxonomy.	140 documents annotated; QA workflow specification drafted; Review pass completed on first 80 documents; 20+ edge cases
Day 10-11	Completion & IAA Study Setup	Complete all 200 document annotations. Finalize the edge case taxonomy. Set up the IAA study. Complete remaining guideline sections.	200 documents annotated; Guideline complete (30+ pages); Edge case taxonomy complete (30+ cases); IAA study initiated

Day	Focus Area	Key Activities	Milestone Checkpoints
Day 12-13	IAA Analysis & Quality Finalization	Complete the IAA study. Compute all agreement metrics. Conduct error analysis. Perform final QA pass on gold-standard dataset. Compile efficiency metrics.	IAA data collection complete; All metrics computed; Error analysis completed; Final QA pass done; Efficiency metrics compiled
Day 14-15	Documentation & Submission	Write the IAA report. Finalize all documentation. Ensure all deliverables conform to the repository structure. Conduct final self-review. Transfer repository to ZethetaIntern.	All 7 deliverables finalized; Repository structure verified; README completed; Repository transferred to ZethetaIntern

Day 1-3 - Foundation & Schema Design

- Read this entire assessment document twice. Highlight all quantitative targets and deadlines.
- Set up GitHub repository with the exact directory structure specified in Section C.2.1. Create stub files for all deliverables.
- Install and configure Label Studio. Create the project with entity types and colour scheme.
- Design the annotation schema (D1). Start with entity types, then add relationships, then validation rules. Use the entity taxonomy in Section A.6 as your starting point.
- Begin sourcing or generating the document corpus. Aim to have at least 50 documents ready for annotation by end of day.
- Make your first commits. Push the schema draft and repository structure. Begin drafting entity type definitions for the guideline.

Day 4-5 - Annotation Guideline Development

- Write Sections 1–4 of the annotation guideline (Introduction, Document Classification Guide, Entity Type Definitions, Annotation Conventions). Aim for 15+ pages.
- Annotate the first 30 documents (pilot batch). Use this experience to validate the schema and identify initial edge cases.
- Revise the schema based on pilot annotation experience. Fix any issues discovered during annotation.
- Commit all progress. Verify schema validates against pilot annotations. Document any edge cases encountered.

Day 6-8 - Edge Cases & Guideline Refinement

- Annotate documents 31–80. Document every edge case as you encounter it. This is prime edge-case-discovery territory.
- Complete Sections 5–6 of the guideline (Edge Case Compendium, Document-Specific Instructions).
- Refine the schema based on 80-document experience. Commit all updates. You should have 15+ edge cases documented.

Day 9 - Annotation Sprint & Quality Gates

- Annotation sprint targeting 60 documents (documents 81–140). This is where your learning curve pays off.
- Draft the QA workflow specification (D6). You now have enough experience to describe a realistic QA process.
- Self-review the first 80 documents. Run validation scripts. Fix any errors discovered.
- Update edge case taxonomy (target 20+ cases). Commit all progress.

Day 10-11 - Completion & IAA Study Setup

- Complete annotations for documents 141–200. Maintain quality despite fatigue.
- Finalize the annotation guideline (all 9 sections). Complete the edge case taxonomy (30+ cases).
- Set up the IAA study. If using Method A, brief peer annotators and share the guideline. If using Method C, configure the LLM annotator with your guideline as the system prompt.
- Run validation on all 200 documents. Begin collecting IAA data (this can run overnight for Method C).

Day 12-13 - IAA Analysis & Quality Finalization

- Complete IAA data collection. Ensure all 30 documents have dual annotations.
- Compute all IAA metrics using your `compute_iaa.py` script. Generate visualisations. Compute confidence intervals.
- Conduct error analysis on all disagreements. Identify top 10 sources. Propose guideline revisions.
- Final QA pass on the gold-standard dataset. Compile efficiency metrics. Run `generate_statistics.py`.

Day 14-15 - Documentation & Submission

- Write the IAA report (D4). This is the most analytically demanding writing task.
 - Finalize all reports (D5 edge cases, D6 QA workflow, D7 efficiency metrics). Complete the README.md.
 - Final commit. Verify repository structure one last time. Run all validation scripts.
 - Transfer repository to ZethetaIntern. Send confirmation email. DO NOT delete local clone.
- C.2 GitHub Repository Structure & Submission Requirements

Mandatory Repository Structure

The GitHub repository must conform exactly to the following directory structure. Deviations will result in point deductions. The assessor will run automated validation scripts against this structure.

File/Directory	Location	Description
README.md	Root	Project overview, setup instructions, deliverable index, and self-assessment summary
schema/annotation_schema.json	schema/	The annotation schema (D1) in JSON Schema format
schema/entity_types.md	schema/	Human-readable entity type reference card
schema/classification_hierarchy.md	schema/	Document classification hierarchy with decision rules
guidelines/annotation_guideline.pdf	guidelines/	The complete annotation guideline (D2) as PDF
guidelines/annotation_guideline_source.md	guidelines/	Source document (Markdown/LaTeX) for the guideline
guidelines/quick_reference_card.pdf	guidelines/	2-page quick reference card for annotators
data/gold_standard/annotation_s.jsonl	data/gold_standard/	The complete gold-standard dataset (one doc per line)
data/gold_standard/statistics.json	data/gold_standard/	Summary statistics of the annotated dataset
data/raw_documents/	data/raw_documents/	Directory containing the original 200 documents (or references)
data/iaa_study/	data/iaa_study/	Raw annotation data from the IAA study

File/Directory	Location	Description
reports/iaa_report.pdf	reports/	The Inter-Annotator Agreement Report (D4)
reports/edge_case_taxonomy.md	reports/	The Edge Case Taxonomy (D5)
reports/qa_workflow.md	reports/	The QA Workflow Specification (D6)
reports/efficiency_metrics.md	reports/	Annotation Efficiency Metrics (D7)
scripts/validate_annotations.py	scripts/	Script to validate annotations against schema
scripts/compute_iaa.py	scripts/	Script to compute IAA metrics
scripts/generate_statistics.py	scripts/	Script to generate dataset statistics
.github/CODEOWNERS	.github/	Code ownership for review purposes
LICENSE	Root	MIT License or as specified

README.md Requirements

The README.md file is the first thing the assessor will read. It must contain:

1. Project title and intern identification (name, email, cohort)
2. Executive summary (200–300 words) describing the approach taken and key outcomes
3. Table of deliverables with completion status and self-assessed quality rating
4. Setup and installation instructions (how to validate annotations, compute IAA metrics)
5. Key design decisions and trade-offs made during the project
6. Known limitations and areas for improvement
7. Time log summary (total hours spent, breakdown by day and activity)

PART E: TOOLS AND PLATFORMS GUIDE

SECTION E1 - REQUIRED TECHNOLOGY STACK

Tool	Purpose	Version
Git / GitHub	Version control, submission	Latest
Label Studio	Annotation platform for document labelling	1.10+
Python 3.10+	Script development (validation, IAA computation)	3.10+
Notion / Markdown Editor	Annotation guideline authoring	Any
Google Sheets / Excel	IAA calculations and data analysis	Any
scikit-learn (Python)	Cohen's Kappa and F1 computation	1.3+
Regex101	Entity pattern validation and testing	Web-based
JSON Schema Validator	Schema validation (ajv, jsonschema)	Any
Pandoc (optional)	Document format conversion	Latest
Prodigy / Doccano (reference)	Reference annotation tools for best practices	Any

SECTION E2 - LABEL STUDIO CONFIGURATION GUIDE

Interns must configure Label Studio with the following project settings. Create a new Label Studio project with the labelling interface XML configuration that maps all 22+ entity types with distinct colour codes for visual differentiation.

E.2.1 Project Template Configuration

The Label Studio interface must be configured with the following structure: a document classification section using Choices (single-select) for the 8 primary document types, a subtype classification using nested Choices for each primary type, a quality tag section using Choices (multi-select) for the 10 quality tags, and an entity annotation section using Labels with at least 22 entity types, each with a distinct background colour for visual identification.

E.2.2 Recommended Entity Colour Scheme

Entity Type	Recommended Colour	Hex Code	Rationale
MONETARY_A MOUNT	Red	#FF6B6B	High visual priority; money is the most frequently annotated entity
DATE	Teal	#4ECDC4	Distinct from monetary; frequently co-occurs with amounts
ACCOUNT_NU MBER	Blue	#45B7D1	Cool colour family; distinct from party names
PARTY_NAME	Green	#96CEB4	Natural colour for 'people/organisations'
REGULATORY_ ID	Yellow	#FFEAA7	Warm highlight colour; stands out in dense text
ADDRESS	Purple	#DDA0DD	Distinct from all other entity families
INTEREST_RAT E	Orange	#F0B27A	Related to monetary amounts but distinct
DURATION	Light Blue	#AED6F1	Time-related; pairs visually with dates
PERCENTAGE	Coral	#F1948A	Related to rates; visually close to monetary

Entity Type	Recommended Colour	Hex Code	Rationale
DOCUMENT_ID	Grey	#AEB6BF	Administrative metadata; lower visual priority
CLAUSE_REFERENCE	Dark Green	#52BE80	Legal/structural; distinct from party names
TRANSACTION_TYPE	Pink	#F5B7B1	Transaction metadata
INSTITUTION_NAME	Dark Teal	#48C9B0	Similar family to party names but distinct
PRODUCT_NAME	Lavender	#BB8FCE	Product/commercial entities
STATUS	Amber	#F7DC6F	Status/state indicators
CONTACT_INFO	Light Grey	#D5DBDB	Contact details; low visual priority
SIGNATURE_BLOCK	Dark Purple	#8E44AD	Signature elements; high formality
TABLE_DATA	Light Green	#ABEBC6	Structural data; large spans
COMPUTATION	Dark Orange	#E67E22	Calculated values; requires attention
FOOTNOTE_REF	Brown	#A04000	Reference markers; small spans
CURRENCY_CODE	Dark Red	#C0392B	Currency identifiers; relates to amounts
FISCAL_PERIOD	Steel Blue	#5DADE2	Time periods; relates to dates

E.2.3 Quality Assurance Settings

Enable the following Label Studio features for quality assurance:

- Review mode: Enable reviewer assignment for all completed annotations
- Agreement metrics: Enable built-in agreement tracking between annotators
- Annotation history: Retain all versions of annotations for audit trail and rollback
- Webhook notifications: Set up notification for review completion (optional but recommended)
- Data export format: Configure default export to JSON format matching your annotation schema
- Prediction integration: If using Method C for IAA, configure a Machine Learning backend for pre-annotation



SECTION E3 - PYTHON ENVIRONMENT SETUP

The following Python packages are required for the validation and IAA computation scripts. Create a requirements.txt with: scikit-learn>=1.3.0, pandas>=2.0.0, numpy>=1.24.0, jsonschema>=4.20.0, matplotlib>=3.7.0, seaborn>=0.12.0, scipy>=1.11.0, krippendorff>=0.6.0, tabulate>=0.9.0, and rich>=13.0.0.



SECTION E4 -ANNOTATION METHODOLOGY - DETAILED TECHNICAL GUIDANCE

E.4.1 The Annotation Lifecycle

Every document in the corpus must pass through the following four-stage annotation lifecycle. This lifecycle is not optional - it represents industry best practice for producing high-quality training data.

Stage 1: Pre-Annotation Assessment

- Read the entire document once to understand its type, structure, and content
- Assign the primary document type and subtype classification
- Assign quality tags (OCR quality, language, currency, etc.)
- Estimate annotation difficulty (1–5 scale) and record in metadata
- Start the time-tracking clock

Stage 2: First-Pass Entity Extraction

- Annotate all clearly identifiable MONETARY_AMOUNT entities
- Annotate all DATE entities with normalised forms
- Annotate all PARTY_NAME entities with role disambiguation
- Annotate all REGULATORY_ID and ACCOUNT_NUMBER entities
- Flag any entities that require second-pass review with annotator notes

Stage 3: Second-Pass Deep Annotation

- Annotate TABLE_DATA entities with row/column structure preservation
- Annotate COMPUTATION entities with formula documentation
- Annotate CLAUSE_REFERENCE entities with hierarchical structure
- Resolve all flagged entities from the first pass
- Establish entity-to-entity RELATIONSHIPS
- Verify normalised values for all entities

Stage 4: Self-Review & Quality Check

- Run the schema validation script against the annotation
- Verify all mandatory entity types have been considered (even if absent from the document)
- Cross-check monetary totals against stated sums where available
- Verify date normalisations are correct (timezone-aware where relevant)
- Record time-to-annotate and stop the clock

E.4.2 Span Selection Rules

One of the most critical annotation decisions is determining the exact span (start and end positions) of each entity. Inconsistent span selection is the single largest source of inter-annotator disagreement. The minimal span principle must be followed:

Text	Correct Span	Incorrect Span	Reason
Total amount due: Rs. 45,230.00	Rs. 45,230.00	Total amount due: Rs. 45,230.00	Minimal span includes currency symbol but not label text
Dated: 15th March 2024	15th March 2024	Dated: 15th March 2024	Minimal span excludes the field label
PAN: ABCDE1234F	ABCDE1234F	PAN: ABCDE1234F	Entity is the ID itself, not the label
Account No. 1234567890123	1234567890123	Account No. 1234567890123	Entity is the number; label provides context
Mr. Rajesh Kumar Singh	Rajesh Kumar Singh	Mr. Rajesh Kumar Singh	Exclude honorifics unless they disambiguate

Exceptions to minimal span: ADDRESS (include complete address), CLAUSE_REFERENCE (include full reference), SIGNATURE_BLOCK (include name, designation, and signing date as single entity), and TABLE_DATA (include entire table structure including headers).

E.4.3 Normalisation Standards

Entity Type	Normalised Form Example	Rules
MONETARY_AMOUNT	{ amount: 45230.00, currency: 'INR' }	Always use ISO 4217 currency code; amounts as decimal numbers; remove thousands separators
DATE	2024-03-15	ISO 8601 format (YYYY-MM-DD); fiscal year dates: FY2023-24; partial dates: 2024-03-XX
ACCOUNT_NUMBER	****7890 (masked)	Last 4 digits visible; full number in encrypted field; format per institution rules
REGULATOR_ID	{ type: 'PAN', value: 'ABCDE1234F' }	Include ID type; validate checksum; uppercase normalisation
INTEREST_RATE	{ rate: 8.50, period: 'annual', type: 'APR' }	Specify period and type; convert to annual equivalent for comparison
DURATION	{ value: 36, unit: 'months' }	Normalise to smallest unambiguous unit; '3 years' becomes 36 months
PERCENTAGE	{ value: 18.00, type: 'GST' }	Include percentage type/context; decimal precision to 2 places
TRANSACTION_TYPE	DEBIT	Map to standard enum: DEBIT, CREDIT, TRANSFER, REVERSAL, ADJUSTMENT, REFUND

SECTION E5 - INTER-ANNOTATOR AGREEMENT - STATISTICAL METHODOLOGY

E.5.1 Cohen's Kappa - Classification Agreement

Cohen's Kappa measures the agreement between two annotators on categorical decisions (document classification), correcting for agreement that would occur by chance. Formula: $\kappa = (P_o - P_e) / (1 - P_e)$, where P_o = observed agreement proportion, P_e = expected agreement by chance.

Kappa Range	Interpretation	Project Implication
0.81 – 1.00	Almost Perfect Agreement	Annotation guideline is highly effective; ready for production use
0.61 – 0.80	Substantial Agreement	Guideline needs minor refinement; most categories are well-defined
0.41 – 0.60	Moderate Agreement	Significant guideline issues; major revision needed before production
0.21 – 0.40	Fair Agreement	Guideline is unclear; fundamental redesign required
0.00 – 0.20	Slight Agreement	Annotations are essentially random; start over
< 0.00	Less Than Chance	Systematic disagreement; annotators may be inverting categories

E.5.2 Entity-Level F1 Score Computation

For entity extraction agreement, the project uses span-based evaluation with both strict and relaxed matching criteria. Strict matching requires identical entity type, start position, and end position. Relaxed matching requires identical entity type and at least 50% span overlap measured by Jaccard similarity.

Computation procedure: (1) Designate one annotator as reference and the other as predicted. (2) For each entity type, count True Positives, False Positives, and False Negatives. (3) Compute Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, $F1 = 2PR / (P+R)$. (4) Compute macro-F1 and micro-F1. (5) Report both strict and relaxed scores.

E.5.3 Statistical Significance Testing

- Bootstrap Confidence Intervals: Resample the document subset 1000 times with replacement, compute Kappa for each sample, report the 95% CI
- Permutation Test: Randomly shuffle annotations between annotators 1000 times, compute the proportion of permuted Kappas exceeding the observed Kappa (this is the p-value)
- Effect Size: Report the difference between observed Kappa and the threshold (0.85) as a standardised effect size



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SECTION E6 - EDGE CASE ANALYSIS FRAMEWORK

E.6.1 Edge Case Identification Protocol

An annotation scenario qualifies as an edge case if: (1) the annotation guideline does not explicitly address the scenario, (2) two reasonable annotators could defensibly annotate it differently, (3) the correct annotation depends on document-specific context, (4) the scenario involves conflicting annotation rules, or (5) the scenario reveals a schema design limitation.

E.6.2 Exemplar Edge Cases

EC-SAMPLE-001: Ambiguous Currency Symbol

Field	Content
Category	Entity Extraction - MONETARY_AMOUNT / CURRENCY_CODE
Difficulty	Level 2 (Uncommon)
Description	The dollar sign (\$) appears in a document without any other contextual clue about which dollar currency is intended
Example Text	'Total Payment: \$12,450.00' - Document header says 'XYZ Trading Pte. Ltd.' (Singapore entity)
Resolution	Infer currency from institutional context. Singapore entity + \$ = SGD. Annotate as {amount: 12450.00, currency: 'SGD'} with confidence: 0.85
Rejected Alternatives	(1) Default to USD: Rejected due to US-centrism. (2) Leave ambiguous: Rejected as downstream needs a code. (3) Annotate both: Rejected - one normalised value required
Frequency	Estimated 8–12 documents out of 200
Model Impact	Incorrect currency assignment would cause 100% error on monetary value in non-USD jurisdictions. Critical severity.

EC-SAMPLE-002: OCR-Corrupted Date

Field	Content
Category	Entity Extraction - DATE / OCR Noise
Difficulty	Level 3 (Rare/Ambiguous)
Description	OCR produces '01/O3/2024' where the actual date is '01/03/2024'. The OCR engine confused 0 with O, 1 with l
Resolution	Apply OCR error correction heuristic. Annotate as DATE: '2024-03-01' with confidence: 0.70. Add OCR_DEGRADED tag.
Frequency	Estimated 15–25 date entities across all scanned documents
Model Impact	Training on uncorrected OCR dates teaches the model to produce invalid formats. Medium-high severity.

EC-SAMPLE-003: Computed Value Discrepancy

Field	Content
Category	Entity Extraction - MONETARY_AMOUNT / COMPUTATION
Difficulty	Level 3 (Rare/Ambiguous)
Description	Invoice states 'Subtotal: 10,000 GST @18%: 1,800 Total: 11,900' but $10,000 + 1,800 = 11,800$, not 11,900
Resolution	Annotate all three amounts as stated. Create a COMPUTATION entity documenting the expected formula. Flag the discrepancy. Do NOT correct the total.
Frequency	Estimated 5–8 documents out of 200
Model Impact	Training to accept arithmetic errors reduces ability to flag incorrect documents. High severity for compliance.

SECTION E7 - QUALITY ASSURANCE WORKFLOW - REFERENCE ARCHITECTURE

E.7.1 Five-Stage QA Pipeline

Stage	Actor	Input	Output	Quality Gate
1. Pre-Screening	QA Lead	Raw document	Screened document with quality tags	Document legibility $\geq 60\%$; critical pages present
2. Primary Annotation	Annotator	Screened document	First-pass annotations	Schema validation passes; all mandatory entity types checked
3. Self-Review	Same Annotator	First-pass annotations	Reviewed annotations	Zero schema errors; entity count within expected range
4. Peer Review	Reviewer	Reviewed annotations	Adjudicated annotations	Reviewer agreement $\geq 90\%$ with annotator on spot-check
5. Final QA	QA Lead	Adjudicated annotations	Gold-standard annotations	Random 10% sample achieves 98%+ accuracy

E.7.2 Escalation Matrix

Issue Type	Severity	Escalation Path	Resolution SLA
Schema doesn't accommodate entity	High	Annotator > QA Lead > Schema Designer	4 hours
Guideline ambiguity	Medium	Annotator > Reviewer > Guideline Author	8 hours
Inter-annotator disagreement	Medium	Annotator > Peer > Adjudicator	12 hours
Document quality below threshold	Low	Annotator > QA Lead (exclude or OCR-correct)	24 hours
Systematic annotation error discovered	Critical	QA Lead > Re-annotate all affected documents	48 hours

E.7.3 Sampling Strategy for Quality Verification

Annotation Phase	Review Sampling Rate	Rationale
First 30 documents (pilot)	100% full review	Calibrate the annotator; identify guideline gaps early
Documents 31–80	50% random sample	Annotator is improving; reduce overhead while maintaining quality
Documents 81–140	30% random sample	Annotator is consistent; focus review on flagged documents
Documents 141–200	20% random + 100% flagged	Annotator is proficient; minimal oversight with targeted review
Final QA pass	10% stratified random sample	Verify overall quality; sample proportional to document categories

PART F: SUBMISSION PROTOCOL

Projects are assessed using a comprehensive 1000-point rubric covering seven dimensions: Problem Understanding, Solution Quality, Research & Analysis, Presentation & Clarity, Innovation & Creativity, Feasibility & Practicality and CV alignment. Assessment is conducted through a combination of manual review and AI-powered evaluation.



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END OF PROJECT DOCUMENT